

Rates and Unit Rates

Lesson 4-1

Name: _____

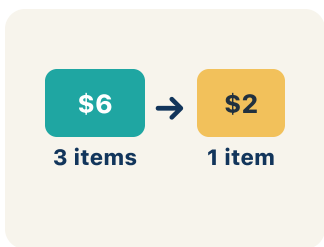
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



\$12 for 4 games is a rate

Rate

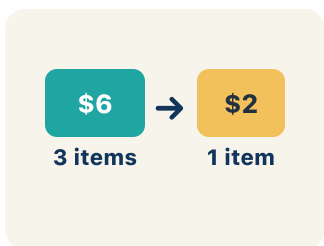
Write the definition:



\$3 per 1 game

Unit Rate

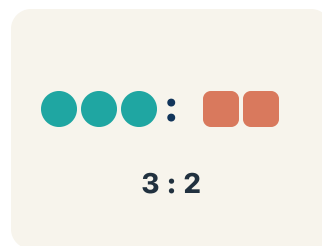
Write the definition:



60 miles per hour means 60 miles for every 1 hour

Per

Write the definition:

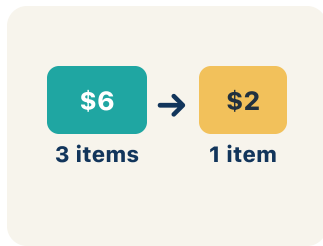


3 : 2

5 to 3 or 5:3

Ratio

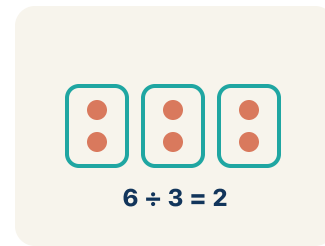
Write the definition:



Pack A: \$0.50 per pencil vs Pack B: \$0.40 per pencil — Pack B is the better buy

Better Buy

Write the definition:



\$12 ÷ 4 games = \$3 per game

Divide

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find a unit rate to compare prices and decide the better buy.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rate

Unit Rate

Per

Ratio

Better Buy

Divide

1 A ratio that compares two quantities with different units is a _____.

2 A rate that tells the cost or amount for one unit is a _____.

3 The word that means 'for each one' is _____.

4 A comparison of two quantities is a _____.

5 The option with the lower unit rate, costing less per item, is the _____.

6 To find a unit rate, I _____ the price by the number of units.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A booth charges \$4.50 for 9 games. What is the unit rate?

- A. \$0.50 per game
- B. \$0.45 per game
- C. \$2.00 per game
- D. \$4.50 per game

Step 1 $\$4.50 \div 9 = \0.50 per game.

 **Answer:** A. \$0.50 per game

 We do — together

Solve this one with your class using the same steps.

Problem: A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?

- A. 30 miles per gallon
- B. 6 miles per gallon
- C. 180 miles per gallon
- D. 36 miles per gallon

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: A pack of 8 markers costs \$6.00. What is the cost per marker?

- A. \$0.75
- B. \$0.80
- C. \$1.33
- D. \$0.60

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Two cabinets give points over time. Nova scored 78 points in 6 minutes. Pulse scored 75 points in 5 minutes. Which cabinet has the faster scoring rate (more points per minute)?

- A. Pulse, at 15 points per minute
- B. Nova, at 13 points per minute
- C. They score at the same rate
- D. There is not enough information

Show your work:

2. Two claw machines cost coins per play. Machine A charges 150 coins for 5 plays. Machine B charges 168 coins for 8 plays. Which machine is the better value (fewer coins per play)?

- A. Machine B, at 21 coins per play
- B. Machine A, at 30 coins per play
- C. They are the same value
- D. Machine A, because it costs fewer coins total

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.

Sentence starter: The student is ____ because the best deal is the ____ pack with a unit rate of \$____ per token.

Show your work:

Reflect — Exit Ticket

A store sells 6 pencils for \$1.50 and 10 pencils for \$2.80. Which is the better buy?

- A. 6 for \$1.50 — \$0.25 each
- B. 10 for \$2.80 — \$0.28 each
- C. They cost the same per pencil
- D. Not enough information

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Rate
2. **Notes 2:** Unit Rate
3. **Notes 3:** Per
4. **Notes 4:** Ratio
5. **Notes 5:** Better Buy
6. **Notes 6:** Divide
7. **We Try (notes):** A. 30 miles per gallon — $180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon}$.
8. **You Try (notes):** A. $\$0.75 - \$6.00 \div 8 = \$0.75 \text{ per marker}$.
9. **Try It 1:** A. Pulse, at 15 points per minute — Nova: $78 \div 6 = 13 \text{ points per minute}$. Pulse: $75 \div 5 = 15 \text{ points per minute}$. $15 > 13$, so Pulse is faster.
10. **Try It 2:** A. Machine B, at 21 coins per play — Machine A: $150 \div 5 = 30 \text{ coins per play}$. Machine B: $168 \div 8 = 21 \text{ coins per play}$. $21 < 30$, so Machine B is the better value.
11. **Exit Ticket:** A. 6 for \$1.50 — \$0.25 each — $\$1.50 \div 6 = \0.25 per pencil . $\$2.80 \div 10 = \0.28 per pencil . $\$0.25 < \0.28 , so 6 for \$1.50 is the better buy.